



Figure 5: SuSE Linux installation could be fun

Control -> 1. Power On/Off System-> 8. Power on. After that, in approximately two minutes, you will hear the noise of the fans—the big system has started initialisation and a minute later the hardware will be completely initialised, signalled by two speaker beeps. At the same time, on the console, you should see a message from SMS (System Management Services) as in Figure 2, where you choose '1' to enter the system OF menu and then select 1. *Select Install/Boot Device* to choose the boot device (which you will run the installer from). You should see something like Figure 3, where I chose 4. *IDE*, because the built-in CD-ROM is connected to the IDE connector. In your version, it might be 3. *CD/DVD*. In the subsequent menu (Figure 4), I chose 2. *Normal Mode Boot*, which is suitable for installation, and confirmed exit from SMS. My SLES 11's Linux kernel correctly identified the IntelliStation 285. The installation process will proceed as follows:

```
Welcome to SuSE Linux Enterprise 11!

Type "install" to start the YaST installer on this CD/DVD
Type "slip"   to start the YaST install via network
Type "rescue" to start the rescue system on this CD/DVD

Welcome to yaboot version r22.8-r1190.SuSE
booted from '/pci@80000002:0000003/pci@2,3/ide@1/disk@0:1,\
suseboot/yaboot.ibm'
running with firmware 'IBM, SF240_403' on model 'IBM,9111-285',
serial 'IBM,0266AB830', partition '0'
Enter "help" to get some basic usage information
boot:
Please wait, loading kernel...
Allocated 01800000 bytes for executable @ 03000000
Elf32 kernel loaded...

SuSE Linux zImage starting: loaded at 03000000-0469c3d0
(4009000/0/02839a68; sp: 02c9fd50)
uncompressing ELF header done. (00000100 bytes)
```



Figure 6: Management via Web interface

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Allocated 0164c090 bytes for kernel @ 04800000
Leave 122a5ba bytes for initrd @ 034668bb
uncompressing kernel done. (00f5a920 bytes)
entering kernel at 04810000(34668bb/122a5ba/02639a68)
```

The installation procedure reminds me of what happens on the x86, with just one difference—in our case, control is via a serial line in text VT102 mode, so some characters may not be displayed properly as you can see in Figure 5.

Several menu items, such as the network address, may require you to return to re-edit these fields. We can use *Tab/Alt+Tab* to move between text interface fields. These keyboard shortcuts are functional even in this very minimal mode.



Note: Installing SLES 10 involved no difficulty in navigation or in selecting the text buttons. However, with SLES 11, there was the problem of having to check the Internet connection (*Test Connection to the Internet Via Ethernet Controller*). The problem is that there is no systems setting to exclude or interrupt this operation when you're in text mode via a serial line. In this case, simply restart the system and assign the necessary IP address after a reboot (via serial connection and *minicom*). After that, you're encouraged to connect via SSH over the Ethernet connection.

Embedded Linux on ServiceProcessor

In addition to control via a serial line, IntelliStation (and in general, all IBM products) offers management via an ordinary Web browser. This Web control mechanism (Advanced System Management—see Figure 6) is provided by a tiny computer within the IntelliStation, with Linux and BusyBox utilities. This ServiceProcessor also uses Power technology—a PowerPC 405 CPU from IBM [5] but with a much smaller form factor. This uses a low-power processor FSP [6] (Flexible Service Processor) running at about 200